

Mert Koca

Mapping neural circuits at nanometer scale, from bench to supercomputer.

LOCATION

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Highly self-driven and intellectually curious individual with a strong foundation in machine learning and neuroscience, working at the interface of expansion-microscopy connectomics and deep-learning segmentation. Adept at synthesizing complex information across disciplines into innovative, strategic solutions, and at managing long-term projects with a holistic approach that balances creative ideation with structured execution.

EXPERIENCE

Jul 2025 - Present

Istanbul, Turkey

Research Intern

KUTTAM, Koç University Research Center for Translational Medicine

FLAGSHIP RESEARCH

- Studied various **Expansion Microscopy (ExM)** protocols (ProExM, Magnify, and single-shot 20× ExM) to surpass the optical diffraction limit, enabling nanometer-scale imaging of neural tissue on confocal microscopes.
- Built and ran deep-learning **volume-segmentation pipelines** (Flood-Filling Networks, U-Net) for neural and vascular structures, working toward computer-based 3D brain mapping.
- Learned and operated **high-performance computing** on TRUBA (Turkey's national HPC center): SLURM batch scheduling, multi-GPU (JAX) training, and Apptainer/Singularity containers.
- This work forms the foundation of the bachelor thesis comparing FFN vs. LSD segmentation methods.*

Oct 2024 - Feb 2026

Scholarship to Jul 2025

Embedded Machine Learning Scholar

TUSEB · Health Institutes of Turkey

- Interpreted **EMG signals** using LSTM+CNN, Random Forest, and Gaussian Process models.
- Developed a real-time signal-interpretation script.
- Produced data-analysis visualizations and animations in Python (Manim).

Aug 2024 - Jul 2025

Start-up

Member

PsyAI

- Built an **agentic parenting-assistant** with LangChain & LangGraph, featuring long/short-term memory and RAG over curated sources.
- Python backend with a UI delivering personalized, framework-based behavioral insights.

Jan 2024 - Jul 2024

National finalist

Team Member

Tez-Türk

- EMG gesture classification** on the Ninapro DB5 dataset; built an MLP classifier for a prosthetic hand.

SELECTED PROJECTS

mertoshi.online *Personal website & agentic RAG platform, solo, end-to-end*

- 🧑‍💻 **Frontend:** React 19 + TypeScript + Vite + Tailwind + Spline 3D, deployed on Azure Static Web Apps.
- 🧑‍💻 **Backend:** Python Azure Functions RAG using Azure OpenAI "On Your Data" + Azure AI Search (hybrid vector/semantic), secured with Managed Identity + Azure Key Vault.
- 🧑‍💻 Features a custom AI assistant ("Lundo") and a document-intelligence + semantic-chunking indexing pipeline.

liconn-ffn-truba *Open-source HPC connectomics pipeline*

- 🧑‍💻 Trains Google's **Flood-Filling Network** (JAX) on real expansion-microscopy connectomics data, scaling 1 → 4 GPUs on TRUBA.
- 🧑‍💻 Public GitHub repository with an animated report.

EDUCATION

B.Sc., Molecular Biotechnology

Turkish-German University · Beykoz, Istanbul · *Graduated*
Jun 2026 · GPA 3.14 / 4.0

Bachelor thesis (in German): light-microscopic connectomics beyond the diffraction limit: deep-learning volume segmentation comparing Flood-Filling Networks (FFN) vs. Local Shape Descriptors (LSD). Advisor: Prof. Dr. Atay Vural.

CERTIFICATION

Foundations of Quantum Mechanics

University of Colorado Boulder · *Coursera*

LANGUAGES

Turkish: Native · **German:** Advanced (C1, medium of instruction) · **English:** Fluent

PUBLICATIONS

- 🧑‍💻 Coskuner-Weber O., Koca M., Uversky V. N. “Molecular Crowding by Computational Approaches.” Springer Nature. DOI: 10.1007/978-3-032-03370-3_21
- 🧑‍💻 Koca M. “Kuantum-Klasik Bilinç Modelleri ve İnterdisipliner Perspektifler.” Quantum Dergisi, Issue 5 (qturkey.org).
- 🧑‍💻 *Third manuscript in preparation.*

SKILLS

PROGRAMMING	Python, TensorFlow, JAX, NumPy, PyTorch (learning), Bash
MACHINE LEARNING	Deep learning (CNN, LSTM, MLP, Flood-Filling Networks, U-Net), Random Forest, Gaussian Processes, LangChain / LangGraph, RAG
HPC & CLOUD	TRUBA HPC, SLURM, multi-GPU training, Apptainer / Singularity, Azure (Functions, AI Search, OpenAI), Git / GitHub
IMAGING & BIO	Expansion microscopy, connectomics, image segmentation, pan-protein staining, immunofluorescence staining
TOOLS & VIZ	Manim, Matplotlib

INTERESTS

Photography (street & portrait) · running · interdisciplinary science · behavioral neuroscience.